

Problem

In a certain application, a simple RC low-pass filter is designed to reduce high frequency noise. If the desired corner frequency is 20 kHz and $C = 0.5 \mu\text{F}$, find the value of R .

Step-by-step solution

Step 1 of 1

$$\omega_c = 2\pi f_c = \frac{1}{RL}$$

$$R = \frac{1}{2\pi f_c C} = \frac{1}{(2\pi)(15)(10 \times 10^{-6})} = 1.061 K\Omega$$